

Project 3: The Penning trap

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We investigate the classical dynamics of charged particles confined in a Penning trap by combining analytical theory and large-scale numerical simulations. Starting from the Lorentz force in an ideal quadrupolar potential and a uniform axial magnetic field, we derive the equations of motion and their solutions, which serve as validation. We then implement two time-integration schemes in C++ (forward Euler and fourth-order Runge-Kutta, RK4) and assess their accuracy against the analytical solution. Beyond a single ion, we study two-particle dynamics with and without Coulomb interaction and analyze phase-space portraits. We also explore driven dynamics for 100 particles under a time-dependent potential $V(t) \propto V_0[1 + f \cos(\omega_v t)]$ by scanning the ω_v frequency and measuring the fraction of trapped particles. These simulations illustrate key features of Penning-trap dynamics relevant for precision experiments.

I. INTRODUCTION

A Penning trap is a device used to store and manipulate charged particles by combining a static electric field with a homogeneous magnetic field. In its idealized form, three electrodes—a ring and two end caps—generate the electrostatic potential, while a solenoidal magnet provides a uniform field along the trap axis (see Fig. 1) [1]. Classical electrodynamics suffices to describe the particle dynamics at the velocities relevant here; radiative and induced magnetic effects are negligible in the electrostatic approximation $\mathcal{O}(v^2/c^2)$ [1].

The electric field $\mathbf{E}$ derives from the scalar potential V via

$$\mathbf{E} = -\nabla V, \quad (1)$$

and, in an ideal Penning trap, the electrodes generate the potential

$$V(x, y, z) = \frac{V_0}{2d^2} (2z^2 - x^2 - y^2), \quad (2)$$

where V_0 is the applied voltage and d is the characteristic trap dimension, defined from the electrode geometry (with $d = \sqrt{z_0^2 + r_0^2}/2$ for hyperbolic end-cap distance z_0 and ring radius r_0) [1]. The magnetic field is taken to be spatially uniform and aligned with the z axis,

$$\mathbf{B} = B_0 \hat{\mathbf{e}}_z, \quad B_0 > 0. \quad (3)$$

The motion of a particle with charge q and mass m follows from Newton's second law under the Lorentz force,

$$m \ddot{\mathbf{r}} = q(\mathbf{E} + \dot{\mathbf{r}} \times \mathbf{B}). \quad (4)$$

Equation (2) provides confinement along z , while the magnetic field suppresses radial escape in the xy -plane [1].

The Penning trap is a versatile tool utilized for highly precise experiments in fundamental physics, providing ultra-precise measurements. By confining charged particles, ranging from elementary particles (such as

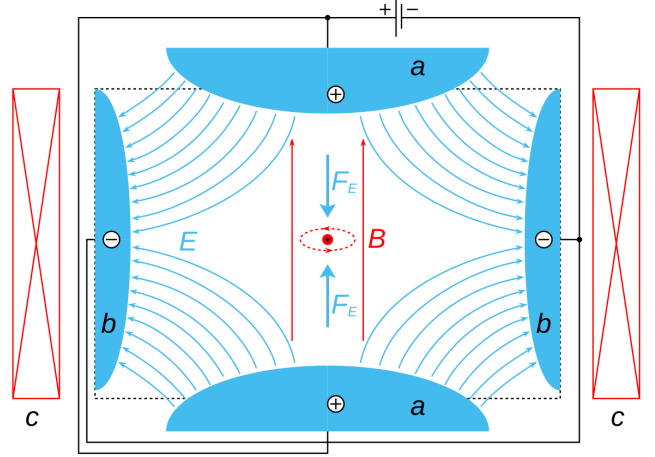


FIG. 1. Schematic of a Penning trap. Two end caps (a), ring electrode (b) and generator of a uniform magnetic field $\mathbf{B}$ (c). The red dot indicates a trapped positively charged particle. [2]

electrons, positrons, protons, and antiprotons) to heavy atoms, using a combination of a strong homogeneous magnetic field and a weak electrostatic field, these devices enable detailed study of basic properties of atomic systems and their constituents [3]. Applications include testing the *Standard Model* of physics, particularly through the precise determination of *fundamental constants* (like the fine-structure constant α and the mass of the electron) [3]. Penning traps also perform stringent tests of *Quantum Electrodynamics (QED)* and the *CPT-invariance theorem* by comparing properties, such as masses and magnetic moments, of particles and antiparticles [3]. Furthermore, Penning trap mass spectrometry is essential for direct mass measurements of exotic nuclides, contributing significantly to advanced problems in nuclear physics and astrophysics, including the study of proton-unstable nuclides, superheavy elements, and rapid proton (*rp*) and rapid neutron (*r*) capture processes [3].

The remainder of this report is organized as follows.

In Sec. II we derive the equations of motion, present the analytical single-particle solutions, and recast the many-particle problem into first-order form suitable for numerical integration. Sec. III details the C++ implementation, including the `Particle` and `PenningTrap` classes and the two integrators (forward Euler and RK4). Sec. IV reports the main results: single-particle validation, two-particle trajectories and phase-space plots with and without Coulomb interaction, accuracy benchmarks (RK4 vs. Euler), and resonance scans for a time-dependent potential. Sec. V summarizes our conclusions and outlines directions for future work. All code developed for this report is available via this GitHub repository.¹

II. METHODS

Equations of motion and general solution

We can write the Cartesian components of eq. (4). For that, we consider eq. (1) and eq. (2) to obtain $E_x = \frac{V_0}{d^2}x$, $E_y = \frac{V_0}{d^2}y$ and $E_z = -\frac{2V_0}{d^2}z$. As the magnetic field satisfies the condition of eq. (3), the cross product in eq. (4) can be written as $\dot{\mathbf{r}} \times \mathbf{B} = (\dot{y}B_0, -\dot{x}B_0, 0)$.

Now, the Cartesian components of eq. (4) are

$$m\ddot{x} = q\left(\frac{V_0}{d^2}x + \dot{y}B_0\right), \quad (5)$$

$$m\ddot{y} = q\left(\frac{V_0}{d^2}y - \dot{x}B_0\right), \quad (6)$$

$$m\ddot{z} = -q\left(\frac{2V_0}{d^2}\right)z. \quad (7)$$

Introducing the standard trap parameters

$$\omega_0 \equiv \frac{qB_0}{m}, \quad \omega_z^2 \equiv \frac{2qV_0}{md^2}, \quad (8)$$

we obtain the equations of motion

$$\ddot{x} - \omega_0\dot{y} - \frac{1}{2}\omega_z^2x = 0, \quad (9)$$

$$\ddot{y} + \omega_0\dot{x} - \frac{1}{2}\omega_z^2y = 0, \quad (10)$$

$$\ddot{z} + \omega_z^2z = 0. \quad (11)$$

As eq. (11) is a simple harmonic oscillator, the general solution for the z component is

$$z(t) = A_z \cos(\omega_z t + \phi_z), \quad (12)$$

with real amplitude A_z and phase ϕ_z fixed by the initial conditions. In the following development, we will assume

$q > 0$.

Equations (9) and (10) are coupled in x and y but can be rewritten to a single differential equation. Defining the complex radial coordinate

$$f(t) \equiv x(t) + iy(t) \quad (13)$$

and adding eq. (9) to i times eq. (10) we obtain,

$$\ddot{f} + i\ddot{y} - \omega_0\dot{y} + i\omega_0\dot{x} - \frac{1}{2}\omega_z^2x - i\frac{1}{2}\omega_z^2y = 0$$

$$\ddot{f} + i\omega_0\dot{f} - \frac{1}{2}\omega_z^2f = 0, \quad (14)$$

From eq. (14) the general solution in the xy -plane can be written as

$$f(t) = A_+ e^{-i(\omega_+ t + \phi_+)} + A_- e^{-i(\omega_- t + \phi_-)}, \quad (15)$$

$$\omega_{\pm} = \frac{\omega_0}{2} \left(1 \pm \sqrt{1 - 2\frac{\omega_z^2}{\omega_0^2}} \right),$$

with real amplitudes $A_{\pm} \geq 0$ and phases $\phi_{\pm}$. A *bounded* radial motion ($|f(t)| < \infty$ as $t \rightarrow \infty$) requires $\omega_{\pm} \in \mathbb{R}$, i.e. the square root to be real. This gives the stability condition

$$\omega_0^2 > 2\omega_z^2 \quad (16)$$

Using the trap parameters introduced in Eq. (8), the condition becomes

$$\left(\frac{qB_0}{m}\right)^2 > 2\left(\frac{2qV_0}{md^2}\right) \iff B_0^2 > \frac{4m}{q} \frac{V_0}{d^2} \iff V_0 < \frac{qB_0^2 d^2}{4m} \quad (17)$$

for $q > 0$ (our working assumption).

From eq. (13) we obtain the radial distance $r_{\perp}(t) = \sqrt{x^2 + y^2} = |f(t)|$. Its square is

$$\begin{aligned} r_{\perp}^2(t) &= f(t)f^*(t) \\ &= A_+^2 + A_-^2 + 2A_+A_- \cos[(\omega_+ - \omega_-)t + (\phi_+ - \phi_-)]. \end{aligned} \quad (18)$$

Since $\cos(\cdot) \in [-1, 1]$, the quantity $r_{\perp}^2$ oscillates between the extrema

$$r_{\perp, \max}^2 = (A_+ + A_-)^2, \quad r_{\perp, \min}^2 = (A_+ - A_-)^2.$$

Therefore the upper and lower bounds on the particle's distance from the origin in the xy -plane are

$$R_+ = A_+ + A_-, \quad R_- = |A_+ - A_-|.$$

¹ <https://github.uio.no/teresm/Comp.Phys.Project/tree/master/P3>

Specific analytical solution for validation

For tests of the Penning trap simulation, we adopt a concrete set of initial conditions for a single particle of charge q and mass m :

$$x(0) = x_0, \quad \dot{x}(0) = 0,$$

$$y(0) = 0, \quad \dot{y}(0) = v_0,$$

$$z(0) = z_0, \quad \dot{z}(0) = 0.$$

From eq. (12) and the initial values for $z(0)$ and $\dot{z}(0)$, we get that the specific solution for $z(t)$ is

$$z(t) = z_0 \cos(\omega_z t). \quad (19)$$

For the movement in the xy -plane, the specific solution for $f(t)$ is given by eq. (15) with

$$A_+ = \frac{v_0 + \omega_- x_0}{\omega_- - \omega_+}, \quad A_- = -\frac{v_0 + \omega_+ x_0}{\omega_- - \omega_+},$$

$$\phi_+ = 0, \quad \phi_- = 0 \quad (20)$$

Many-particle extension and first-order form

Up to now we have considered a single particle. If the trap is filled with n particles of charges $\{q_1, \dots, q_n\}$ and masses $\{m_1, \dots, m_n\}$, each particle experiences the external Penning-trap fields *and* the Coulomb force from all the others. Denoting particle indices by i and j , the coupled equations of motion become

$$\ddot{x}_i - \omega_{0,i} \dot{y}_i - \frac{1}{2} \omega_{z,i}^2 x_i - k_e \frac{q_i}{m_i} \sum_{j \neq i} q_j \frac{x_i - x_j}{|\mathbf{r}_i - \mathbf{r}_j|^3} = 0, \quad (21)$$

$$\ddot{y}_i + \omega_{0,i} \dot{x}_i - \frac{1}{2} \omega_{z,i}^2 y_i - k_e \frac{q_i}{m_i} \sum_{j \neq i} q_j \frac{y_i - y_j}{|\mathbf{r}_i - \mathbf{r}_j|^3} = 0, \quad (22)$$

$$\ddot{z}_i + \omega_{z,i}^2 z_i - k_e \frac{q_i}{m_i} \sum_{j \neq i} q_j \frac{z_i - z_j}{|\mathbf{r}_i - \mathbf{r}_j|^3} = 0, \quad (23)$$

where $\mathbf{r}_i = (x_i, y_i, z_i)$, $\omega_{0,i} \equiv q_i B_0 / m_i$, and k_e is Coulomb's constant.

For the numerical treatment we recast these second-order equations as a coupled first-order system by introducing:

$$\dot{\mathbf{r}}_i = \mathbf{v}_i, \quad \dot{\mathbf{v}}_i = \frac{\mathbf{F}_i}{m_i}, \quad (24)$$

where $\mathbf{F}_i$ is the total force acting on a particle at position $\mathbf{r}$ with velocity $\mathbf{v}$ at time t . This formulation connects directly to the single-particle results above and will be the basis for the simulations implemented.

III. CODE DEVELOPMENT

Units and default parameters

For the numerical work that follows we adopt a convenient set of base units: length in micrometres (μm), time in microseconds (μs), mass in atomic mass units (u), and charge in the elementary charge (e). In this system the Coulomb constant is

$$k_e = 1.38935333 \times 10^5 \frac{\text{u} \mu\text{m}^3}{\mu\text{s}^2 e^2},$$

and the SI-derived units convert to

$$1 \text{ T} = 9.64852558 \times 10^1 \frac{\text{u}}{\mu\text{s} e},$$

$$1 \text{ V} = 9.64852558 \times 10^7 \frac{\text{u} \mu\text{m}^2}{\mu\text{s}^2 e}$$

[1].

Our reference trap uses a homogeneous field

$$B_0 = 1.00 \text{ T} \simeq 9.65 \times 10^1 \frac{\text{u}}{\mu\text{s} e},$$

an electrode voltage

$$V_0 = 25.0 \text{ mV} \simeq 2.41 \times 10^6 \frac{\text{u} \mu\text{m}^2}{\mu\text{s}^2 e},$$

and characteristic dimension $d = 500 \mu\text{m}$.

Since the equations only depend on V_0/d^2 , we will frequently use the compact value

$$\frac{V_0}{d^2} = 9.65 \frac{\text{u}}{\mu\text{s}^2 e}.$$

The particles of the simulation are singly charged calcium ions, Ca^+ , which have charge $q = 1e$ and $m = 40.1 \text{ u}$.

Class-oriented design: Particle and PenningTrap

To mirror the dynamical variables and parameters introduced in Section II, we use two minimal C++ classes.

Particle. Stores the properties of one particle: charge q , mass m , position $\mathbf{r}$ and velocity $\mathbf{v}$. In code, $\mathbf{r}$ and $\mathbf{v}$ are `arma::vec` objects.

PenningTrap. Represents the trap configuration and population: magnetic field strength B_0 , applied potential V_0 , characteristic dimension d , and a `std::vector<Particle>` holding all particles. **PenningTrap** includes member functions that provide the external electric field $\mathbf{E}(\mathbf{r})$, the magnetic field $\mathbf{B}(\mathbf{v})$,

the force on particle i from particle j , the total force on particle i from the external fields and the total force on particle i from all the other particles. There is a ultimate member function in `PenningTrap` that uses all the other member functions to compute the total force actin on particle i .

Time integration: Forward Euler and RK4

To evolve the first-order system in eq. (24), we implemented two methods; fourth-order Runge-Kutta (RK4) and forward Euler method. These two procedures are coded within two member functions of the `PenningTrap` class: `evolve_forward_Euler(h)` and `evolve_RK4(h)`. Both advance all particles simultaneously over time using the trap's field/force routines. RK4 is our primary integrator while Forward Euler is included as a simple baseline for verification.

Algorithms 1 and 2 show the pseudocode for the two methods. Here $\mathbf{r}_i$ and $\mathbf{v}_i$ are the position and velocity of particle i , m_i its mass, $\mathbf{F}_i(\{\mathbf{r}\}, \{\mathbf{v}\})$ the total force, and h the time step. Stage vectors are denoted $\mathbf{k}_{r,s}$ and $\mathbf{k}_{v,s}$ ($s = 1, \dots, 4$), each a 3D vector per particle.

Tools

All simulations of the Penning trap have been developed in C++, making use of several standard and scientific software packages. The numerical computations, time evolution algorithms, and data generation were implemented using the following C++ libraries: `<string>`, `<iostream>`, `<vector>`, `<cmath>`, and the `Armadillo` linear algebra package for efficient vector and matrix operations. These tools provided the necessary functionality to construct and evolve the physical system, store particle data, and manage the simulation workflow efficiently.

For the visualization of the results, small `Python` scripts were written to read the output data and generate plots. The main packages employed for this purpose were `NumPy`, used for data handling and numerical processing, and `Matplotlib`, used to create publication-quality figures of particle trajectories, phase-space plots, and resonance scans.

Finally, the writing and revision of the present report have been assisted by `ChatGPT`, which was used as a support tool to improve the clarity, accuracy, and structure of the scientific text.

Algorithm 1 Fourth-order Runge-Kutta (RK4)

$\mathbf{r}_i^{(0)} \leftarrow \mathbf{r}_i, \mathbf{v}_i^{(0)} \leftarrow \mathbf{v}_i$	▷ Copy current state
$\mathbf{F}_i^{(1)} \leftarrow \mathbf{F}_i(\mathbf{r}_i^{(0)}, \mathbf{v}_i^{(0)})$	▷ Compute forces
$\mathbf{k}_{r,1,i} \leftarrow h \mathbf{v}_i^{(0)};$	
$\mathbf{k}_{v,1,i} \leftarrow h \mathbf{F}_i^{(1)}/m_i$	
$\mathbf{r}_i^{(A)} \leftarrow \mathbf{r}_i^{(0)} + \frac{1}{2} \mathbf{k}_{r,1,i}$	▷ Midpoint state A
$\mathbf{v}_i^{(A)} \leftarrow \mathbf{v}_i^{(0)} + \frac{1}{2} \mathbf{k}_{v,1,i}$	
$\mathbf{F}_i^{(2)} \leftarrow \mathbf{F}_i(\mathbf{r}_i^{(A)}, \mathbf{v}_i^{(A)})$	▷ Compute forces
$\mathbf{k}_{r,2,i} \leftarrow h \mathbf{v}_i^{(A)};$	
$\mathbf{k}_{v,2,i} \leftarrow h \mathbf{F}_i^{(2)}/m_i$	
$\mathbf{r}_i^{(B)} \leftarrow \mathbf{r}_i^{(0)} + \frac{1}{2} \mathbf{k}_{r,2,i}$	▷ Midpoint state B
$\mathbf{v}_i^{(B)} \leftarrow \mathbf{v}_i^{(0)} + \frac{1}{2} \mathbf{k}_{v,2,i}$	
$\mathbf{F}_i^{(3)} \leftarrow \mathbf{F}_i(\mathbf{r}_i^{(B)}, \mathbf{v}_i^{(B)})$	▷ Compute forces
$\mathbf{k}_{r,3,i} \leftarrow h \mathbf{v}_i^{(B)};$	
$\mathbf{k}_{v,3,i} \leftarrow h \mathbf{F}_i^{(3)}/m_i$	
$\mathbf{r}_i^{(C)} \leftarrow \mathbf{r}_i^{(0)} + \mathbf{k}_{r,3,i}$	▷ Endpoint state
$\mathbf{v}_i^{(C)} \leftarrow \mathbf{v}_i^{(0)} + \mathbf{k}_{v,3,i}$	
$\mathbf{F}_i^{(4)} \leftarrow \mathbf{F}_i(\mathbf{r}_i^{(C)}, \mathbf{v}_i^{(C)})$	▷ Compute forces
$\mathbf{k}_{r,4,i} \leftarrow h \mathbf{v}_i^{(C)};$	
$\mathbf{k}_{v,4,i} \leftarrow h \mathbf{F}_i^{(4)}/m_i$	
$\mathbf{r}_i \leftarrow \mathbf{r}_i^{(0)} + \frac{1}{6} (\mathbf{k}_{r,1,i} + 2\mathbf{k}_{r,2,i} + 2\mathbf{k}_{r,3,i} + \mathbf{k}_{r,4,i}),$	
$\mathbf{v}_i \leftarrow \mathbf{v}_i^{(0)} + \frac{1}{6} (\mathbf{k}_{v,1,i} + 2\mathbf{k}_{v,2,i} + 2\mathbf{k}_{v,3,i} + \mathbf{k}_{v,4,i}).$	
▷ Update (for all i)	

Algorithm 2 Forward Euler method

$\mathbf{v}_i^{\text{old}} \leftarrow \mathbf{v}_i, \mathbf{r}_i^{\text{old}} \leftarrow \mathbf{r}_i$	▷ Copy velocities and positions
$\mathbf{F}_i \leftarrow \mathbf{F}_i(\mathbf{r}_i^{\text{old}}, \mathbf{v}_i^{\text{old}})$	▷ Compute forces at current state
$\mathbf{v}_i \leftarrow \mathbf{v}_i^{\text{old}} + h \mathbf{F}_i/m_i$	▷ Velocity update
$\mathbf{r}_i \leftarrow \mathbf{r}_i^{\text{old}} + h \mathbf{v}_i^{\text{old}}$	▷ Position update

IV. RESULTS AND DISCUSSION

Single particle dynamics in the Penning trap

A single charged particle has been simulated inside the Penning trap in order to verify the correctness of the implemented numerical methods. The parameters used for

the simulation correspond to a Ca^+ ion, whose physical and initial conditions are summarized below:

- **Particle:** mass $m = 40.1 \text{ u}$, charge $q = 1 \text{ e}$, initial position $\mathbf{r}_0 = (20, 0, 20) \mu\text{m}$, and initial velocity $\mathbf{v}_0 = (0, 25, 0) \frac{\mu\text{m}}{\mu\text{s}}$.
- **Penning trap:** magnetic field $B_0 = 9.65 \times 10^1 \frac{\text{u}}{\mu\text{s} \text{ e}}$, electric potential $V_0 = 2.41 \times 10^6 \frac{\text{u} \mu\text{m}^2}{\mu\text{s}^2 \text{ e}}$, and characteristic dimension $d = 500 \mu\text{m}$.

The simulation was carried out over a total time of $50 \mu\text{s}$ using 1000 time steps of the fourth-order Runge–Kutta (RK4) integration method. Figure 2 shows the resulting trajectory of the axial coordinate $r_z(t)$.

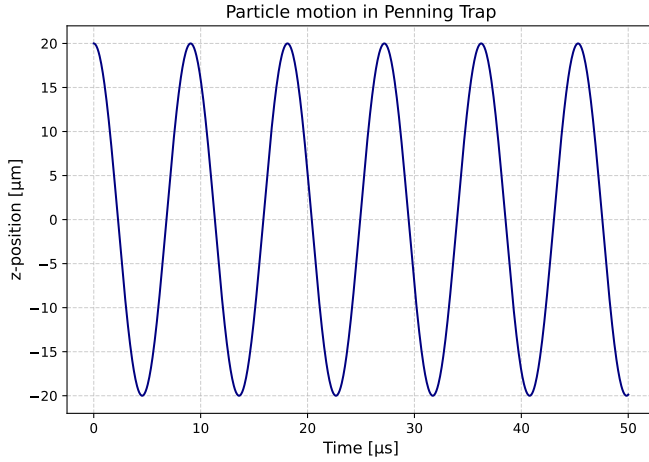


FIG. 2. Temporal evolution of the axial position r_z of a single Ca^+ ion confined in the Penning trap.

The $r_z(t)$ curve exhibits a clear oscillatory motion with an amplitude of approximately $\pm 20 \mu\text{m}$, in agreement with the analytical solution for the axial component given by eq. (19). This confirms that the implementation of the electromagnetic fields and the RK4 time-evolution method correctly reproduce the expected harmonic motion along the z -direction, where only the electric field acts on the ion.

Two-particle dynamics in the xy -plane

To further explore the system behavior, a simulation with two charged particles was performed in the Penning trap. The trap parameters were kept the same as in the single-particle case. Two singly-charged calcium ions (Ca^+) were introduced into the trap. The first ion was initialized at position $\mathbf{r}_1 = (20, 0, 20) \mu\text{m}$ with velocity $\mathbf{v}_1 = (0, 25, 0) \frac{\mu\text{m}}{\mu\text{s}}$, while the second ion was initialized at $\mathbf{r}_2 = (25, 25, 0) \mu\text{m}$ with velocity $\mathbf{v}_2 = (0, 40, 5) \frac{\mu\text{m}}{\mu\text{s}}$.

The simulation was carried out for a total duration of $50 \mu\text{s}$ using the fourth-order Runge–Kutta (RK4) method

with 1000 iterations. Two separate scenarios were investigated: one without Coulomb interactions between the particles and one including those interactions. In Figures 3 and 4 $\mathbf{r}_1$ and $\mathbf{v}_1$ correspond to particle (a) whereas $\mathbf{r}_2$ and $\mathbf{v}_2$ correspond to particle (b).

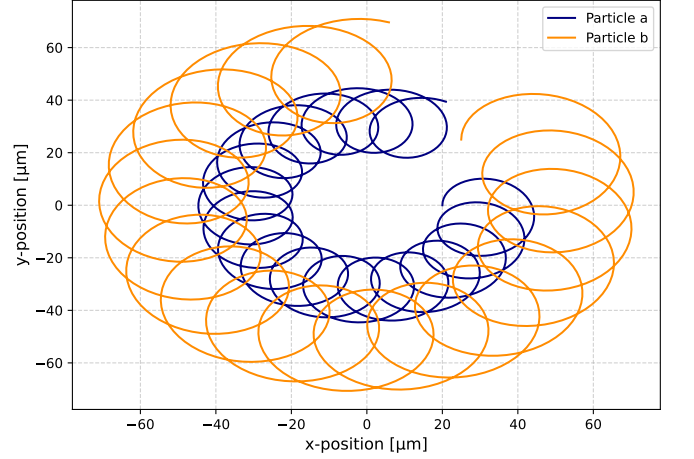


FIG. 3. Trajectories of the two Ca^+ ions in the xy -plane without Coulomb interaction.

In the first case, without the inter-particle Coulomb forces, both particles move independently under the influence of the external electric and magnetic fields. Their trajectories in the xy -plane describe characteristic circular orbits caused by the Lorentz force. The difference in the initial conditions (mainly in the velocity magnitude and direction) results in one particle describing a larger and faster orbit than the other.

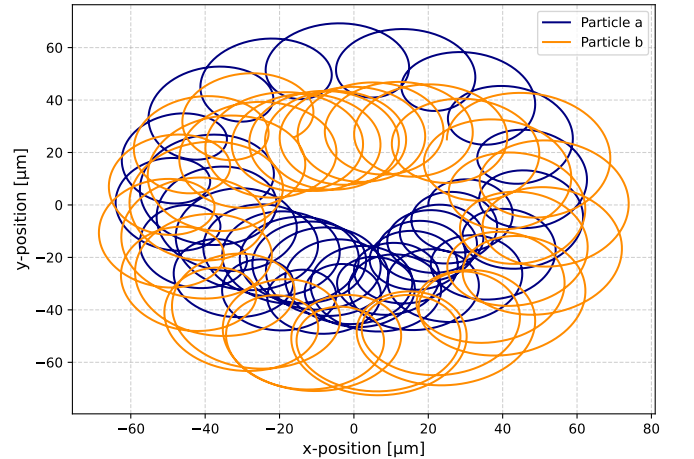


FIG. 4. Trajectories of the two Ca^+ ions in the xy -plane including Coulomb interaction.

When the Coulomb interaction between the ions is included, the motion becomes more complex. The electrostatic repulsion between the positively charged particles modifies their trajectories, breaking the circular symme-

try observed in the non-interacting case. The ions tend to move apart from each other, and their orbits deviate from purely circular paths due to the mutual perturbations induced by the repulsive force.

Phase space analysis: (r_x, v_x) and (r_z, v_z) plots

Continuing with the two-particle simulation under the same initial conditions described previously, we now analyze the phase space plots for both ions. Specifically, the (r_x, v_x) and (r_z, v_z) projections are presented for the cases with and without Coulomb interaction between the particles.

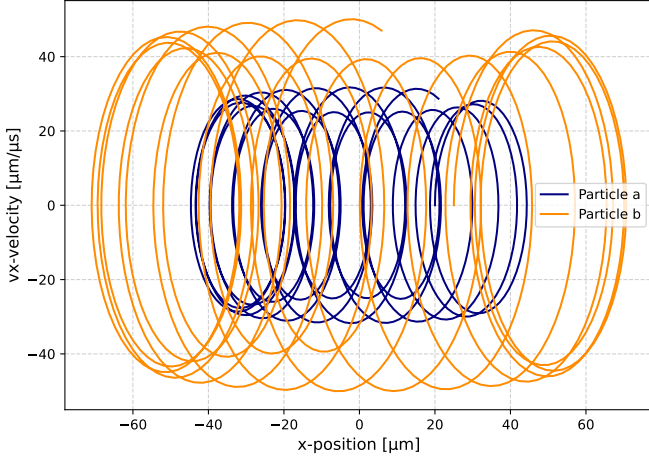


FIG. 5. Phase space plot (r_x, v_x) for both Ca^+ ions without Coulomb interaction.

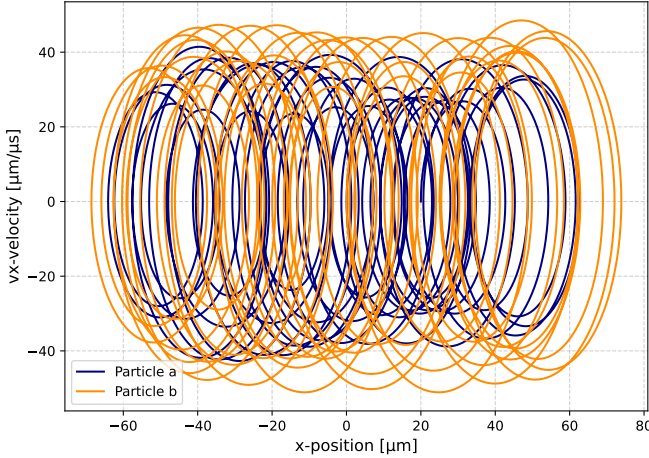


FIG. 6. Phase space plot (r_x, v_x) for both Ca^+ ions including Coulomb interaction.

In the (r_x, v_x) plot for the non-interacting case (see Figure 5), the same periodic motion pattern along the x -axis can be identified as in Figure 3. In both plots,

the maximum amplitude of the movement along this axis is approximately $\pm 42 \mu\text{m}$ for particle (a) and about $\pm 65 \mu\text{m}$ for particle (b). When Coulomb interactions are included, the semi-periodic structure becomes distorted and the motion is perturbed (see Figure 6), consistent with the deviation observed in Figure 4.

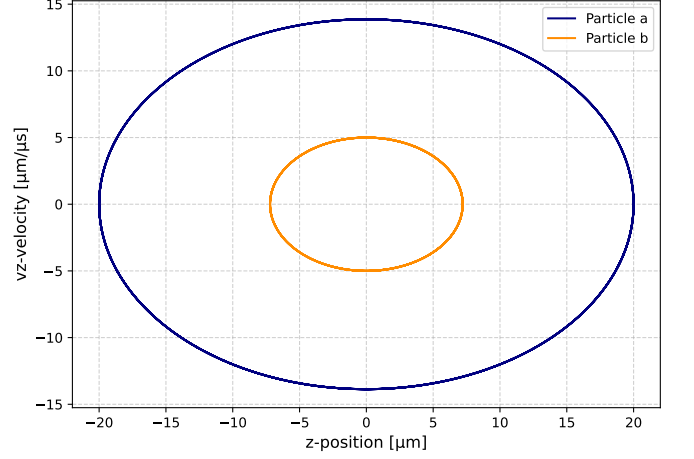


FIG. 7. Phase space plot (r_z, v_z) for both Ca^+ ions without Coulomb interaction.

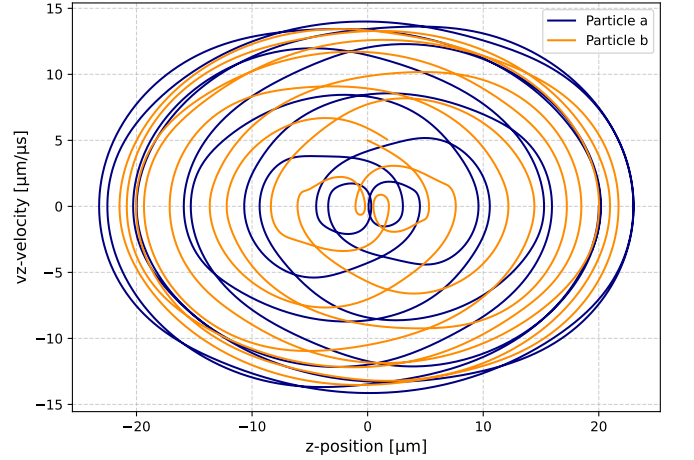


FIG. 8. Phase space plot (r_z, v_z) for both Ca^+ ions including Coulomb interaction.

In the (r_z, v_z) phase space plots for the non-interacting particles (see Figure 7), the expected oscillatory motion along the z -axis is clearly observed, as defined by equation (12). The amplitude of the oscillation ranges between $\pm 20 \mu\text{m}$ for particle (a) and approximately $\pm 7.2 \mu\text{m}$ for particle (b). The case of particle (a) matches the motion presented in Figure 2, since it corresponds to the same particle under identical initial and boundary conditions. The motion of particle (b) also follows equation (12), but with an initial condition where the position along z is zero and the velocity is

non-zero, introducing a phase shift ϕ_z .

When Coulomb interaction is considered (see Figure 8), the oscillatory motion of both particles becomes distorted due to the mutual repulsion, resulting in non-symmetric trajectories in the (r_z, v_z) phase space.

Accuracy assessment: RK4 vs. forward Euler for a single particle

We return to the single Ca^+ ion with initial conditions $(\mathbf{r}_1, \mathbf{v}_1)$ and compare the accuracy of the fourth-order Runge-Kutta (RK4) method (see algorithm 1) against the forward Euler method (see algorithm 2). For a time grid $\{t_i\}$ we quantify the discrepancy using the relative error

$$\varepsilon_i = \frac{\|\mathbf{r}_{\text{exact}}(t_i) - \mathbf{r}(t_i)\|}{\|\mathbf{r}_{\text{exact}}(t_i)\|}, \quad (25)$$

where $\mathbf{r}(t_i)$ is the numerical position at t_i and $\mathbf{r}_{\text{exact}}(t_i)$ is obtained from the analytical solution derived in Sec. II (see eqs. (13), (15) and (19)). We repeated the simulation for $N_{\text{steps}} \in \{4000, 8000, 16000, 32000\}$ with both methods and plotted $\varepsilon(t)$ (see Figures 9 and 10).

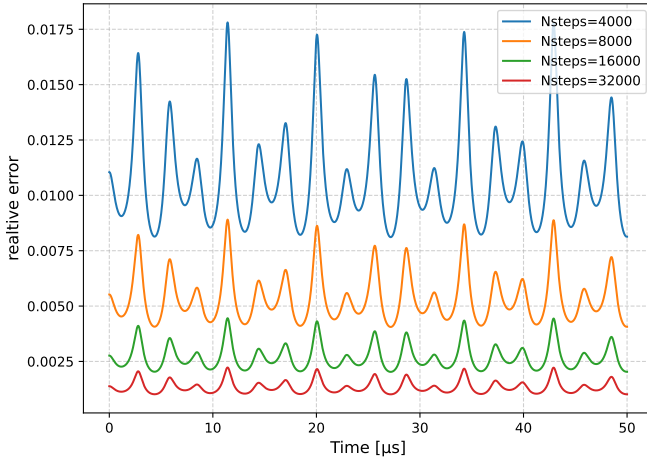


FIG. 9. Relative error $\varepsilon(t)$ for the RK4 integrator at different time resolutions N_{steps} .

Beyond the numerical evidence, the difference between the curves in Figs. 9–10 follows directly from the orders of accuracy of the two integrators. Forward Euler is a first-order method: its local truncation error is $\mathcal{O}(h^2)$ and the global error scales as $\mathcal{O}(h)$. In contrast, RK4 is fourth-order, with local truncation error $\mathcal{O}(h^5)$ and global error $\mathcal{O}(h^4)$. For a given small step h , the factor h^4 is much smaller than h , so RK4 achieves several orders of magnitude lower relative error. This also means that, for a target accuracy, RK4 tolerates larger time steps than Euler.

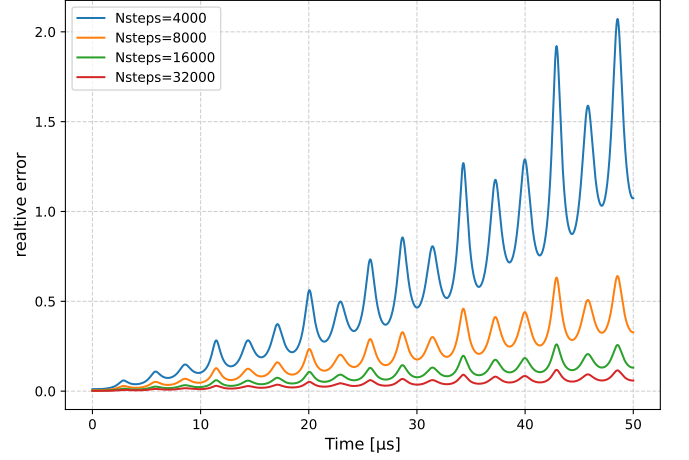


FIG. 10. Relative error $\varepsilon(t)$ for the forward Euler integrator at different time resolutions N_{steps} .

The distinct time evolution of the relative error is tied to stability and error accumulation. Forward Euler does not preserve geometric (e.g. energy) properties in oscillatory/Hamiltonian problems and therefore suffers from error drift: small local biases compound in time, leading to a steadily increasing $\varepsilon(t)$. RK4 is not symplectic either, but its much smaller local errors strongly limit the propagation of phase and amplitude defects, keeping the global error bounded over the simulated interval. In both schemes, refining the time grid (increasing N_{steps} and thus decreasing h) systematically reduces the error, with a far steeper convergence for RK4 in agreement with its fourth-order global accuracy.

Resonance scan with a time-dependent potential

To probe resonance phenomena in our Penning trap, we modulate the applied potential as

$$V(\mathbf{r}, t) = \frac{1}{2d^2} (2z^2 - x^2 - y^2) V_0 (1 + f \cos(\omega_v t)).$$

The trap is populated with $N_0 = 100$ singly charged Ca^+ ions whose initial positions and velocities are sampled from zero-mean normal distributions scaled to the trap size parameter d . For each drive amplitude $f \in \{0.1, 0.4, 0.7\}$ we sweep the angular frequency over $\omega_v \in [0.2, 2.5]$ MHz in steps of 0.02 MHz. The equations of motion are integrated for 500 μs using a fourth-order Runge-Kutta (RK4) scheme. Outside the trap region, defined by $|\mathbf{r}| > d$, both the external electric and magnetic fields are set to zero, so particles leaving this domain cease to experience trap forces. Coulomb interactions between ions are switched off during this scan to isolate single-particle resonances and reduce computational cost.

Figure 11 shows a characteristic resonance structure: regions of high trapping efficiency are interrupted by

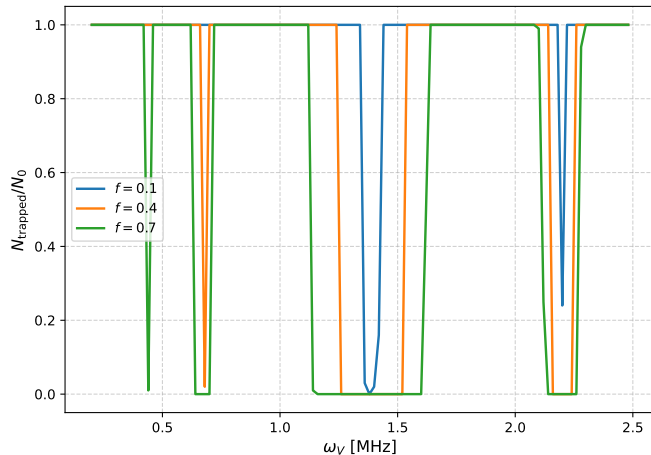


FIG. 11. Fraction of ions that remain trapped after $500 \mu\text{s}$ as a function of drive frequency ω_v for modulation amplitudes $f = 0.1$ (blue), 0.4 (orange), and 0.7 (green). Each curve starts from $N_0 = 100$ randomly initialized Ca^+ ions; Coulomb interactions are disabled.

narrow frequency bands where a significant fraction of ions is lost. These dips occur near the trap eigenfrequencies (axial ω_z and radial $\omega_{\pm}$) or simple combinations thereof. Increasing the modulation amplitude from $f = 0.1$ to 0.7 strengthens the coupling to the motion, broadening and deepening the resonances and reducing the overall trapped fraction across a wider portion of the scan.

To obtain a more detailed view of the resonance behavior, the simulation was repeated for a narrower frequency range, $\omega_v \in [1.0, 1.75]$ MHz, with smaller increments of 0.01 MHz. This refinement effectively provides a zoom into the resonance region identified in the previous broad scan. The corresponding results for the fraction of trapped particles, N_{trapped}/N_0 , are shown in Fig. 12.

In this finer scan, the structure of the main resonance becomes clearly visible. Each curve exhibits a well-defined dip centered near $\omega_v \approx 1.38$ MHz, that reflects a sharp loss in confinement. The width of the resonance again increase with the modulation amplitude f , demonstrating stronger coupling between the oscillating potential and the ion motion as the perturbation amplitude grows.

To study how particle–particle interactions modify this resonance, the same zoomed frequency scan was repeated while including Coulomb forces among the ions. The resulting curves are displayed in Fig. 13.

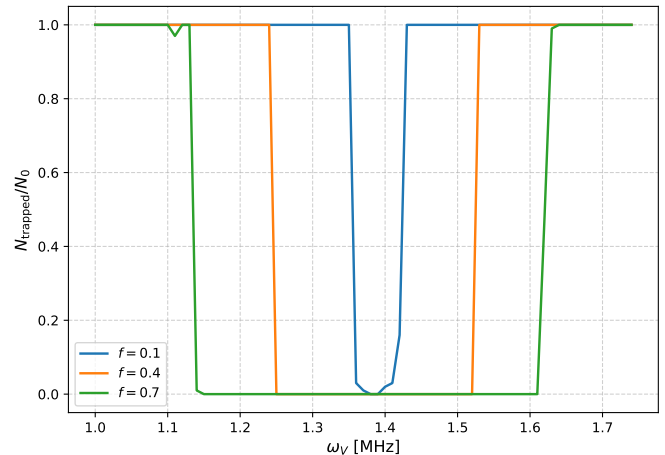


FIG. 12. Zoomed scan of the trapped fraction N_{trapped}/N_0 as a function of ω_v in the range $[1.0, 1.75]$ MHz, for modulation amplitudes $f = 0.1$, 0.4 , and 0.7 , without Coulomb interactions.

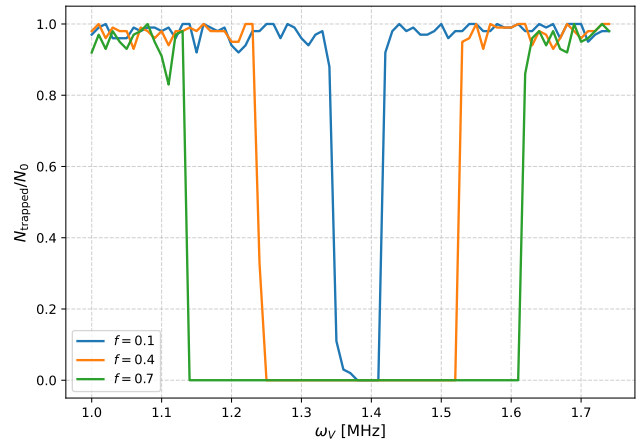


FIG. 13. Zoomed frequency scan including Coulomb interactions between particles. The same range $[1.0, 1.75]$ MHz and amplitudes $f = 0.1$, 0.4 , and 0.7 are used as in the non-interacting case.

When Coulomb interactions are enabled, the overall trapping efficiency decreases across the entire frequency range. We see noise in the regions near the resonance frequency because mutual repulsion between ions adds additional perturbations to their trajectories and introduces dephasing between individual oscillations. This collective effect is expected to smooth out the narrow single-particle resonance structure, leading to a more gradual reduction in the trapped fraction as the driving frequency approaches resonance. The comparison between Figs. 12 and 13 could represent the situation more accurately if a finer frequency spectrum were chosen, that is, using smaller increments of ω_v and a larger number of time steps in the RK4 evolution. However, this is computationally expensive as 100 particles with their respective

interactions are required.

V. CONCLUSION

We have developed and validated a C++ framework to simulate charged-particle motion in an ideal Penning trap. Starting from first principles, we derived the governing equations, implemented forward Euler and RK4 time integrators, and verified the code by comparing single-particle trajectories against exact analytical solutions. We then examined coupled dynamics for two ions, highlighting how Coulomb repulsion perturbs otherwise regular radial motion, and we mapped the response of many non-interacting ions to a time-dependent drive $V(t) = V_0[1 + f \cos(\omega_v t)]$, observing clear resonance-induced loss bands whose width and depth grow with f . Across all tests, RK4 consistently outperformed forward Euler, yielding substantially lower relative errors for the same step size.

We found that (i) RK4 provides stable and accurate trajectories over experimentally relevant times, whereas forward Euler exhibits growing errors; (ii) interparticle interactions break circular symmetry and broaden phase-space structures; and (iii) driving near trap eigenfrequencies leads to significant reductions in the trapped particle fraction, with features that sharpen (or broaden) as the f amplitude decreases (or increases).

For future improvements, several extensions can be considered. First, the implementation of the particle-system evolution algorithm could be optimized to improve computational efficiency and reduce execution time. Second, the resonance-scan results for N_{trapped}/N_0 could be refined by performing a more detailed frequency sweep, using smaller increments in ω_v to resolve the resonance structure with higher precision. Finally, it would be interesting to introduce a periodic time dependence in the external magnetic field and investigate how this modulation affects the fraction of particles that remain confined within the trap over time.

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- [1] Anders Kvellestad. Project 3. <https://anderkve.github.io/FYS3150/book/projects/project3.html#practicalities>, 2025. Accessed on October 4, 2025.
 - [2] Wikipedia contributors. Penning Trap. https://en.wikipedia.org/wiki/File:Penning_Trap.svg, 2025. Accessed on October 4, 2025.
 - [3] K. Blaum, Yu. N. Novikov, and G. Werth. Penning traps as a versatile tool for precise experiments in fundamental physics. *Contemporary Physics*, 51(2):149–175, 2010.